

Subject: Abitatha Roy's Statement of Purpose for Amazon MLSS 2026

Machine learning, for me, has always been something I work with rather than study in the abstract. My B.Tech. program at Cluster Innovation Centre, University of Delhi, combines Information Technology with Mathematical Innovations — a structure that pushed me early into quantitative reasoning alongside implementation. That combination has shaped how I approach every project I take on.

In my research internship at CIC, I developed a feature-selection framework for AQI forecasting using CPCB air-quality datasets spanning four Indian cities, benchmarking ensemble methods — CatBoost, XGBoost, LightGBM, Random Forest — and achieving R^2 up to 0.96 with compact predictor subsets. A separate project involved predicting tuberculosis treatment failure using a stacked ensemble of XGBoost, MLP, and Logistic Regression on a highly imbalanced clinical dataset, where the core challenge was designing a source-aware pipeline that was honest about recall limitations rather than optimising for a misleading accuracy figure. Both manuscripts are under preparation. My team is also presenting a paper at the International Marketing Conference 2026 at IIM Jammu, wherein PLS-SEM modelling and clustering was applied to evaluate AI-mediated consumer-brand relationships — a different domain, but the same discipline of choosing the right model for the question rather than the most impressive one.

Currently, I am interning at the Enterprise Survey Division of the Ministry of Statistics and Programme Implementation in Kolkata, analysing paradata from the ASISSE survey and proposing anomaly detection frameworks using statistical methods and unsupervised learning to identify atypical enterprise response patterns. The work is ongoing and the scope is open-ended, which I find is where I learn the most.

What I want from Amazon ML Summer School is structured depth. My experience so far has been project-driven and applied — I have implemented, evaluated, and iterated, but largely self-directed and within the scope of what each project demanded. What I have not had is systematic exposure to the breadth of ML as Amazon's scientists practice it: the theoretical grounding behind the methods I use, the design decisions that make models production-ready, and the perspective of working at a scale where the cost of a wrong assumption is not a paper revision but a real-world outcome. That is what MLSS offers, and it is precisely what my current trajectory needs.

My mathematical foundation covers linear algebra, probability and statistics, and calculus — the core of the MLSS syllabus. I program primarily in Python, with regular use of Scikit-learn, XGBoost, TensorFlow/Keras, and Pandas. I am prepared for the selection test and, more importantly, for the program itself.

I am applying because I want to learn from people who build ML systems at scale — and because I am at exactly the point in my development where that kind of learning will compound most.